The Rational Polynomials: From Axioms to Factorization

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Contents

1. Introduction

The integers are powerful because their arithmetic has a deep structure: we can divide with remainder, use Euclid's Algorithm to find greatest common divisors, and prove that prime factorization is unique. This paper builds the same story for rational polynomials in Q[x]. Starting from the basic properties of the integers and rationals, we construct Q[x], define a norm based on degree, and use that norm to prove a division algorithm for polynomials. From there, the familiar integer machinery returns like Euclid's Algorithm gives gcds, Bezout's Identity writes them as linear combinations, and Euclid's Lemma explains how prime polynomials divide products. These results come together to prove unique factorization in Q[x].

2. Integers

2.1. Ring and Order Axioms

The integers $\mathbb{Z}$ have addition and multiplication satisfying two sets of axioms.

Definition 1 (Ring axioms)**.** The following axioms define a **Ring**.

- $a + b \in \mathbb{Z}$ and $ab \in \mathbb{Z}$ for all $a, b \in \mathbb{Z}$.
- $a + b = b + a$ and $ab = ba$.
- $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
- $a(b + c) = ab + ac$.
- There is an additive identity $0 \in \mathbb{Z}$ such that $a + 0 = a$.
- There is a multiplicative identity $1 \in \mathbb{Z}$ such that $a \cdot 1 = a$.
- For every $a \in \mathbb{Z}$, there is $-a \in \mathbb{Z}$ such that $a + (-a) = 0$.

Definition 2 (Order axioms)**.** Let $\mathbb{Z}^+$ be the set of positive integers. The following axioms define an **Ordered Ring**

- If $a, b \in \mathbb{Z}^+$, then $a + b \in \mathbb{Z}^+$.
- If $a, b \in \mathbb{Z}^+$, then $ab \in \mathbb{Z}^+$.
- $0 \notin \mathbb{Z}^+$.
- For every $a \in \mathbb{Z}$, exactly one of $a \in \mathbb{Z}^+$, $a = 0$, or $-a \in \mathbb{Z}^+$ is true.

To fully define the integers, our final axiom is:

Definition 3 (Well-Ordering Principle)**.** Every nonempty subset of $\mathbb{Z}^+$ has a least element.

Definition 4. For $a, b \in \mathbb{Z}$, define $a < b$ if $b - a \in \mathbb{Z}^+$. Define $a > b$ if $b < a$. Define $a \leq b$ if $a < b$ or $a = b$, and define $a \geq b$ if $a > b$ or $a = b$.

2.2. Basic Arithmetic

Lemma 2.1. $0 \neq 1$.

Proof. By trichotomy, exactly one of $1 \in \mathbb{Z}^+$, $1 = 0$, or $-1 \in \mathbb{Z}^+$ is true. If $-1 \in \mathbb{Z}^+$, then by multiplicative closure $(-1)(-1) \in \mathbb{Z}^+$. Since $(-1)(-1) = 1$, this gives $1 \in \mathbb{Z}^+$, contradicting trichotomy. Thus $-1 \notin \mathbb{Z}^+$. Also 1 cannot equal 0, since then $-1 = 0$, which would again contradict trichotomy. Hence $1 \in \mathbb{Z}^+$, so $1 \neq 0$. $\square$

Lemma 2.2. *If $a + b = a + b'$, then $b = b'$.*

Proof. Add $-a$ to both sides. Then $(-a)+(a+b) = (-a)+(a+b')$. By associativity, $((-a)+a)+b = ((-a) + a) + b'$, so $0 + b = 0 + b'$. Hence $b = b'$. $\square$

Lemma 2.3. *For all $a, b \in \mathbb{Z}$, $a \cdot 0 = 0$, $-(-a) = a$, $(-a)b = -(ab)$, and $(-a)(-b) = ab$.*

Proof. First, $a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0$. By additive cancellation, $a \cdot 0 = 0$.

Next, since $a + (-a) = 0$, the element a is an additive inverse of $-a$. By uniqueness of additive inverses, $-(-a) = a$.

Also, $ab + (-a)b = (a + (-a))b = 0b = 0$, so $(-a)b$ is the additive inverse of ab. Thus $(-a)b = -(ab)$. Finally, $(-a)(-b) = -\big(a(-b)\big) = -(-(ab)) = ab$. $\square$

Lemma 2.4. *For $a, b, c \in \mathbb{Z}$, if $a \leq b$ and $b \leq c$, then $a \leq c$.*

Proof. If one of the inequalities is equality, the result is immediate. Now suppose $a < b$ and $b < c$. Then $b-a \in \mathbb{Z}^+$ and $c-b \in \mathbb{Z}^+$. By additive closure, $(b-a)+(c-b) \in \mathbb{Z}^+$. But $(b-a)+(c-b) = c-a$, so $c - a \in \mathbb{Z}^+$. Thus $a < c$, and hence $a \leq c$. $\square$

Lemma 2.5. *For $a, b \in \mathbb{Z}$, exactly one of $a < b$, $a = b$, or $b < a$ is true.*

Proof. Apply trichotomy to $b - a$. Exactly one of $b - a \in \mathbb{Z}^+$, $b - a = 0$, or $-(b - a) \in \mathbb{Z}^+$ is true. These are exactly $a < b$, $a = b$, and $b < a$. $\square$

Lemma 2.6. *If $a \leq b$ and $x \leq y$, then $a + x \leq b + y$.*

Proof. If both inequalities are equalities, the result is clear. Otherwise, the differences $b - a$ and $y - x$ are each either positive or zero. Their sum is therefore positive or zero. But $(b+y)-(a+x) = (b - a) + (y - x)$, so $a + x \leq b + y$. $\square$

Lemma 2.7. *If $a, b \in \mathbb{Z}^+$, then $ab \geq a$.*

Lemma 2.8 (NIBZO)**.** *There is no integer strictly between 0 and 1. Equivalently, if $n \in \mathbb{Z}$ and*

$$0 < n < 1,$$

then such an n cannot exist.

Proof. Since $b \in \mathbb{Z}^+$, NIBZO gives $1 \leq b$. If $b = 1$, then $ab = a$. If $1 < b$, then $b - 1 \in \mathbb{Z}^+$. Since $a \in \mathbb{Z}^+$, we get $a(b-1) \in \mathbb{Z}^+$. But $a(b-1) = ab - a$, so $a < ab$. Hence $a \leq ab$. $\square$

Lemma 2.9. *If $a, b \in \mathbb{Z}^+$, then $ab = a$ if and only if $b = 1$.*

Proof. If $b = 1$, then $ab = a$. Conversely, suppose $ab = a$. If $b \neq 1$, then since $b \in \mathbb{Z}^+$ and $1 \leq b$, we have $1 < b$. Thus $b - 1 \in \mathbb{Z}^+$. Since $a \in \mathbb{Z}^+$, $a(b-1) \in \mathbb{Z}^+$. But $a(b-1) = ab - a = 0$, contradicting $0 \notin \mathbb{Z}^+$. Therefore $b = 1$. $\square$

Proof. Suppose, for contradiction, that there is an integer strictly between 0 and 1. Put

$$S = \{n \in \mathbb{Z} : 0 < n < 1\}.$$

Then S is nonempty and $S \subseteq \mathbb{Z}_{\geq 0}$. By the well-ordering principle, S has a least element. Call it m.

Since $0 < m$, multiplicative closure of positive integers gives $0 < m^2$. Since $m < 1$ and $m > 0$, multiplying the inequality $m < 1$ by the positive integer m gives

$$m^2 < m.$$

Thus m^2 is also an integer satisfying

$$0 < m^2 < m < 1.$$

So $m^2 \in S$, but $m^2 < m$. This contradicts the choice of m as the least element of S. Therefore S is empty. $\square$

Definition 5. Define **summation**, or $\sum_{i=i_0}^{n} a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\sum_{i=i_0}^{i_0} = a_{i_0} \qquad \sum_{i=i_0}^{n} a_i = a_n + \sum_{i=i_0}^{n-1} a_i$$

Definition 6. Define **product**, or $\prod_{i=i_0}^{n} a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\prod_{i=i_0}^{i_0} = a_{i_0} \qquad \prod_{i=i_0}^{n} a_i = a_n \cdot \prod_{i=i_0}^{n-1} a_i$$

For properties of products and summations, see Appendix A.

Definition 7. We define exponentiation a^n for $a \in R$, where R is some unital ring, and $n \in \mathbb{Z}^+ \cup \{0\}$ to be:

- 1 if $n = 0$
- $a^{n-1}a$ if $n > 0$

For properties of exponents, see Appendix B.

Definition 8 (Divisibility)**.** For $a, b \in R$, where R is some unital ring, we say a divides b, written

$$a \mid b,$$

if there exists $q \in \mathbb{Q}[x]$ such that

$$b = aq.$$

3. Rationals

In order to construct rational polynomials, we first must formally define rationals.

3.1. Construction

Define $\mathbb{Q} \subseteq \mathbb{Z} \times \mathbb{Z}$, where the elements of $\mathbb{Q}$ are of the form (a, b) where $a, b \in \mathbb{Z}, b \neq 0$. Define two operations $+, \cdot$, such that

- $+ : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ where $(a, b) + (c, d) = (ad + bc, bd)$
- $\cdot : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ where $(a, b) \cdot (c, d) = (ac, bd)$

Also, define an equivalence relation on $\mathbb{Q}$. Two elements $(a, b), (c, d)$ in $\mathbb{Q}$ are equal iff $ad = bc$. Denote equivalence as $(a, b) \sim (c, d)$.

We can also denote (a, b) as $\frac{a}{b}$. The rationals are a Ring as defined by the ring axioms. For $a, b, c, d, e, f \in \mathbb{Z}$,

3.2. Ring Axioms

Commutativity of Multiplication:

$$(a,b)\cdot(c,d) = (ac,bd) \qquad \text{and} \qquad (c,d)\cdot(a,b) = (ca,db) = (ac,bd)$$

Commutativity of Addition:

$$(a,b)+(c,d) = (ad+bc,bd) \qquad \text{and} \qquad (c,d)+(a,b) = (cb+ad,db) = (ad+bc,bd)$$

Associativity of Addition:

$$((a,b)+(c,d))+(e,f) = (a,b)+((c,d)+(e,f))$$

For the left hand side,

$$(ad+bc,bd)+(e,f) = (adf+bcf+bde,bdf)$$

For the right hand side

$$(a,b)+(cf+de,df) = (adf+bcf+bde,bdf).$$

Associativity of Multiplication:

$$(a,b)\cdot((c,d)\cdot(e,f)) = ((a,b)\cdot(c,d))\cdot(e,f)$$

For the left hand side,

$$(a,b)\cdot(ce,df) = (ace,bdf)$$

For the right hand side,

$$(ac,bd)\cdot(e,f) = (ace,bdf).$$

Additive Identity:

$$(a,b)+(0,1) = (a,b)$$

From the definition, $(a,b)+(0,1)$ equals

$$(a\cdot 1 + b\cdot 0, b\cdot 1)$$

By the one axiom and zero axiom of $\mathbb{Z}$,

$$(a,b)$$

Multiplicative Identity:

$$(a, b) \cdot (1, 1) = (a, b)$$

From the definition,

$$(a, b) \cdot (1, 1)$$

equals

$$(a \cdot 1, b \cdot 1)$$

By the one axiom of $\mathbb{Z}$,

$$(a, b)$$

Distributivity: For $x, y, z \in \mathbb{Q}$, we have $x(y + z) = xy + xz$.

Proof. Suppose $x = (a, b), y = (c, d), z = (e, f)$. Then

$$\begin{aligned} x(y + z) &= (a, b) \cdot ((c, d) + (e, f)) = (a, b) \cdot (cf + de, df) \\ &= (a(cf + de), b(df)) \\ &= (acf + ade, bdf) \end{aligned}$$

For the other side,

$$xy = (a, b) \cdot (c, d) = (ac, bd) \qquad \text{and} \qquad xz = (a, b) \cdot (e, f) = (ae, bf)$$

We then have $xy+xz = (ac, bd)+(ae, bf) = (acbf+bdae, bdbf) = (acf+dae, dbf) = (acf+ade, bdf)$. Hence $x(y + z) = xy + xz$. Right distributivity follows immediately, since $(y + z)x = x(y + z) = xy + xz = yx + zx$.

□

3.3. Equivalence

Two rationals are equal if and only if they can be simplified to each other. In other words,

$$(a, b) \sim (c, d) \Leftrightarrow ad = bc$$

Symmetry: If $(a, b) \sim (c, d)$, then $(c, d) \sim (a, b)$

Proof. $ad = bc$. This can be written as $cb = da$, so

$$(c, d) \sim (a, b)$$

$\square$

Transitivity: If $(a, b) \sim (c, d), (c, d) \sim (e, f)$ then $(a, b) \sim (e, f)$.

Proof. Since $ad = bc, cf = de$, $adcf = bcde$ and $cdaf = cdbe$. Thus, $af = be$ so $(a, b) \sim (e, f)$ $\square$

3.4. Well-Defined

To show that addition is well defined, we show that $(a, b) + (c, d) = (a, b) + (e, f)$ if $(c, d) \sim (e, f)$. According to our criterion for an equivalence relation, $cf = de$. Therefore, we want to show $(ad+bc, bd) = (af+be, bf)$. To show these are equivalent, we want to show $abdf+b^2ed = abdf+b^2cf$. We cancel $abdf$ and b^2 because b is nonzero to show that $de = cf$, which is true by our equivalence relation.

To show that multiplication is well defined, we show that $(a, b) \cdot (c, d) = (a, b) \cdot (e, f)$ if $(c, d) \sim (e, f)$. According to our criterion for an equivalence relation, $cf = de$. Therefore, we want to show $(ac, bd) = (ae, bf)$. To show these are equivalent, we want to show $acbf = bdae$. Suppose $a = 0$. Then $0 = 0$, which is true. Otherwise, suppose $a \neq 0$. Then because $ab = ac \implies b = c$ when $a \neq 0$, we can cancel ab from both sides to get $cf = de$, which is true by the equivalence $(c, d) = (e, f)$. Hence multiplication is well defined.

3.5. Multiplicative Inverse

Theorem 1. *For any $(a, b) \in \mathbb{Q}$, and $a \neq 0$, there exists a (c, d) where $(a, b)(c, d) = (1, 1)$*

Consider (b, a) ,and $(b, a) \in \mathbb{Q}$ since $a \neq 0$. Then,

$$(a, b)(b, a) = (ab, ba)$$

and

$$(ab, ba) = (ab, ab) \sim (1, 1)$$

so (b, a) suffices as such (c, d).

4. The Ring $\mathbb{Q}[x]$

Now we can formally define our rational polynomial.

4.1. Construction

Definition 9 (The polynomial ring $\mathbb{Q}[x]$)**.** We define a polynomial $f(x) \in \mathbb{Q}[x]$ with degree n if it can be written as

$$f(x) = \sum_{i=0}^{n} a_i x^i$$

where $n \in \mathbb{Z}_{\geq 0}$ and each coefficient a_i lies in $\mathbb{Q}$ and $a_n \neq 0$. For convenience, $f(x)$ can also be notated as f.
Define two operations $+, \cdot$ such that, given $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{m} b_i x^i$

- $+$: $f(x) + g(x) = \sum_{i=0}^{\max(n,m)} (a_i + b_i) x^i$ with $b_i = 0$ for $i > m$ and $a_i = 0$ for $i > n$.
  Since $\mathbb{Q}$ is closed under addition, each $a_i + b_i$ is rational. Hence $f(x) + g(x) \in \mathbb{Q}[x]$.

- $\times$: By Multiplied Sums, $f(x) \cdot g(x) = \sum_{i=0}^{n} \sum_{j=0}^{m} a_i b_j x^{i+j}$

4.2. Ring Axioms

The rational polynomials also follow the ring axioms.
Commutativity of Addition: Addition is commutative because each coefficient is added in $\mathbb{Q}$, and addition in $\mathbb{Q}$ is commutative.
Associativity of Addition: Addition is associative because each coefficient is added in $\mathbb{Q}$, and addition in $\mathbb{Q}$ is associative.
Commutativity of Multiplication: Multiplication is commutative because the coefficient of x^k in $f(x)g(x)$ comes from all products $a_i b_j$ with $i + j = k$, while the coefficient of x^k in $g(x)f(x)$ comes from all products $b_j a_i$ with $j + i = k$. These are the same products, since multiplication in $\mathbb{Q}$ is commutative.
Associativity of Multiplication: Multiplication is associative because the coefficient of x^k in $(f(x)g(x))h(x)$ comes from all products

$$(a_i b_j) c_r$$

with $i + j + r = k$, while the coefficient of x^k in $f(x)(g(x)h(x))$ comes from all products

$$a_i (b_j c_r)$$

with $i + j + r = k$. These are equal because multiplication in $\mathbb{Q}$ is associative.
Distributivity: Distributivity holds coefficient by coefficient. Put missing coefficients of $g(x)$ and $h(x)$ equal to 0. Then the coefficient of x^k in $f(x)(g(x) + h(x))$ uses terms of the form

$$a_i (b_j + c_j).$$

Since

$$a_i(b_j + c_j) = a_ib_j + a_ic_j$$

in $\mathbb{Q}$, we get

$$f(x)(g(x) + h(x)) = f(x)g(x) + f(x)h(x).$$

Additive Identity: The zero polynomial, $0_{\mathbb{Q}[x]} = 0$, is the additive identity, as

$$f(x) + 0_{\mathbb{Q}[x]} = (a_0 + 0) + \sum_{i=1}^{n} a_ix^i$$

Negatives: Define $-f(x)$ as:

$$-f(x) = \sum_{i=0}^{n}(-a_i)x^i$$

This is the additive inverse of $f(x)$ because $a_i + (-a_i) = 0$ for every i.
Multiplicative Identity: Define $1_{\mathbb{Q}[x]} = 1$ as the multiplicative identity. In the product $1_{\mathbb{Q}[x]}f(x)$, the coefficient of x^i is $1 \cdot a_i = a_i$. Hence

$$1_{\mathbb{Q}[x]}f(x) = f(x).$$

4.3. Units

Just like integer rings, the polynomials have units as well that follow the same definition.

Definition 10. Define $u(x) \in \mathbb{Q}[\text{x}]$ as a **unit** if there exists $v(x) \in \mathbb{Q}[\text{x}]$ s.t. $u(x)v(x) = 1$. The set of units in $\mathbb{Q}[\text{x}]$ is denoted $\mathbb{Q}[\text{x}]^\times$.

Lemma 4.1. $u(x) \in \mathbb{Q}[\text{x}]^\times \Leftrightarrow u(x) \in \mathbb{Q}$

Proof. First, prove $u(x) \in \mathbb{Q} \Rightarrow u(x) \in \mathbb{Q}[\text{x}]^\times$. This is a collorary of the Multiplicative Inverse property of $\mathbb{Q}$. Second, prove $u(x) \in \mathbb{Q}[\text{x}]^\times \Rightarrow u(x) \in \mathbb{Q}$. Define $u(x) = \sum_{i=0}^{n} a_ix^i$ and $v(x) = \sum_{j=0}^{m} b_jx^j$, with n and m the degrees of u and v respectively, then

$$u(x)v(x) = \sum_{i=0}^{n}\sum_{j=0}^{m} a_ib_jx^{i+j}$$

Notice that $i + j = m + n$ only once. However, a_mb_n are both nonzero, so $a_mb_n \neq 0$. Thus, we have a nonzero term $a_mb_nx^{m+n} \neq 1$. □

Notice that because units in $\mathbb{Q}[\text{x}]$ are constants, every polynomial has an equivalent one that is monic.

4.4. The Norm

We need a method to define size of a polynomial. The natural choice is its degree; however, this is not multiplicative, and norms are more helpful when they are multiplicative.

Definition 11. For $f(x) \in \mathbb{Q}[x]/\{0\}$, let the **norm**, or $\|f(x)\|$ be equal to

$$\|f(x)\| = 2^{\deg f(x)}$$

Define $\|0\|$ to be 0. with the following properties:

- $\|f(x)\| \cdot \|g(x)\| = \|f(x)g(x)\|$

  Proof. If $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{j=0}^{m} b_j x^j$, then

  $$f(x)g(x) = \sum_{i=0}^{n} \sum_{j=0}^{m} a_i b_j x^{i+j}$$

  Notice that to find the degree of $f(x)g(x)$, we take the maximum value of $i + j$ which is their respective maximums, $n + m$, so:

  $$\|f(x)g(x)\| = 2^{n+m} = 2^n 2^m = \|f(x)g(x)\|$$

  □

- For any $c(x) \in \mathbb{Q}[x]^\times$, $\|c(x)\| = 1$.

  Proof. Since $\deg c(x) = 0$, $\|c(x)\| = 2^0 = 1$. □

- $\|f(x)\| \in \mathbb{Z} \cup \{0\}$.

  Proof. Since $2 \in \mathbb{Z}^+$, 2^n for $n \in \mathbb{Z}^+$ is always positive by Appendix B, the norm $\in \mathbb{Z} \cup \{0\}$ □

We want units (and only units) to preserve norm.

Lemma 4.2. $\|f(x)g(x)\| = \|f(x)\| \Leftrightarrow g(x) \in \mathbb{Q}[\mathrm{x}]^\times$

Proof.
- $g(x) \in \mathbb{Q}[\mathrm{x}]^\times \Rightarrow \|f(x)g(x)\| = \|f(x)\|$

  Since $\|g(x)\| = 1$, $\|f(x)g(x)\| = \|f(x)\| \cdot 1 = \|f(x)\|$.

- $\|f(x)g(x)\| = \|f(x)\| \Rightarrow g(x) \in \mathbb{Q}[\mathrm{x}]^{\times}$
  Since $\|f(x)g(x)\| = \|f(x)\|\|g(x)\|$. As $\|f(x)\|, \|g(x)\| \in \mathbb{Z}^{+}$, if $\|f(x)g(x)\| = \|f(x)\|$, $\|g(x)\| = 1$. Thus, $g(x) \in \mathbb{Q}[\mathrm{x}]^{\times}$.

□

Lemma 4.3. *Let $f(x), g(x) \in \mathbb{Q}[x]$ with $f(x) \neq 0$ and $g(x) \neq 0$. Then*

$$\|f(x)\| \leq \|f(x)g(x)\|.$$

Proof. Since $f(x)g(x) = \sum_{i=0}^{n} \sum_{j=0}^{m} a_i b_j x^{i+j}$, the $\deg(f(x)g(x)) = m + n$ since that is the largest i and j. Therefore

$$\|f(x)g(x)\| = 2^{\deg(fg)} = 2^{m+n}.$$

Since $n \geq 0$,

$$2^m \leq 2^{m+n}.$$

So

$$\|f(x)\| = 2^m \leq 2^{m+n} = \|f(x)g(x)\|.$$

□

In addition, $\mathbb{Q}$[x] is an integral domain.

Theorem 2 ($\mathbb{Q}[x]$ has no zero divisors)**.** *If $f(x), g(x) \in \mathbb{Q}[x]$ and*

$$f(x)g(x) = 0,$$

then $f(x) = 0$ or $g(x) = 0$.

Proof. Suppose, for contradiction, that

$$f(x)g(x) = 0$$

with $f(x) \neq 0$ and $g(x) \neq 0$.

Since $g(x) \neq 0$, by the norm lemma,

$$\|f(x)\| \leq \|f(x)g(x)\|.$$

Since $f(x) \neq 0$, $\|f(x)\|$ is positive. So $\|f(x)g(x)\|$ is positive. In particular, $f(x)g(x) \neq 0$, since the norm is only positive for nonzero polynomials.

This contradicts $f(x)g(x) = 0$. Therefore at least one of $f(x)$ and $g(x)$ must be zero. □

Corollary 3 (Cancellation)**.** *If* $h(x) \neq 0$ *and*

$$h(x)f(x) = h(x)g(x),$$

then $f(x) = g(x)$.

Proof. Subtract the right side from the left:

$$h(f - g) = 0.$$

Since $h \neq 0$ and $\mathbb{Q}[x]$ has no zero divisors, we must have $f - g = 0$. Thus $f = g$. □

4.5. Basic Arithmetic

Almost all the integer properties have analogous properties in $\mathbb{Q}$[x].

Lemma 4.4 (Uniqueness of zero and one)**.** *The additive identity and multiplicative identity in* $\mathbb{Q}[x]$ *are unique. In particular, if* $f(x) + z(x) = f(x)$ *for every* $f(x) \in \mathbb{Q}[x]$, *then* $z(x) = 0$. *Similarly, if* $f(x)u(x) = f(x)$ *for every* $f(x) \in \mathbb{Q}[x]$, *then* $u(x) = 1$.

Proof. Suppose z acts like an additive identity for f, so $f + z = f$. Add $-f$ to both sides:

$$(-f) + (f + z) = (-f) + f.$$

By associativity and the definition of additive inverse, this becomes

$$0 + z = 0.$$

Since 0 is the additive identity, $z = 0$.

Now suppose u acts like a multiplicative identity for every polynomial. Test this on the constant polynomial 1:

$$1 \cdot u = 1.$$

But $1 \cdot u = u$. Hence $u = 1$. □

Lemma 4.5. $0 \neq 1$

Proof. $0_{\mathbb{Q}[x]}$ has constant coefficient 0, while $1_{\mathbb{Q}[x]}$ has constant coefficient 1. Since $0 \neq 1$ in $\mathbb{Q}$,

$$0_{\mathbb{Q}[x]} \neq 1_{\mathbb{Q}[x]}.$$

□

Lemma 4.6 (Uniqueness of additive inverses)**.** *If $u(x)$ and $v(x)$ are both additive inverses of $f(x)$, then $u(x) = v(x)$.*

Proof. Assume

$$f + u = 0 \qquad \text{and} \qquad f + v = 0.$$

Then $f + u = f + v$. Add $-f$ to both sides:

$$(-f) + (f + u) = (-f) + (f + v).$$

By associativity,

$$((-f) + f) + u = ((-f) + f) + v.$$

Thus

$$0 + u = 0 + v,$$

so $u = v$. □

Lemma 4.7 (Zero and negatives)**.** *For all $f(x), g(x) \in \mathbb{Q}[x]$,*

$$f(x) \cdot 0 = 0, \qquad -(-f(x)) = f(x), \qquad (-f(x))(-g(x)) = f(x)g(x).$$

Proof. First,

$$f \cdot 0 = f(0 + 0) = f \cdot 0 + f \cdot 0.$$

Adding the additive inverse of $f \cdot 0$ to both sides gives $f \cdot 0 = 0$.

Next, since

$$(-f) + f = 0,$$

the polynomial f is an additive inverse of $-f$. By uniqueness of additive inverses,

$$-(-f) = f.$$

For the product of negatives, first observe that

$$fg + (-f)g = (f + (-f))g = 0 \cdot g = 0.$$

Therefore $(-f)g = -(fg)$. Applying this with g replaced by $-g$ gives

$$(-f)(-g) = -(f(-g)).$$

Also,

$$fg + f(-g) = f(g + (-g)) = f \cdot 0 = 0,$$

so $f(-g) = -(fg)$. Hence

$$(-f)(-g) = -(-(fg)) = fg.$$

$\square$

4.6. Divisibility

The greatest common divsor is also helpful in many theorems.

Definition 12 (Greatest Common Divisor)**.** For $f(x), g(x) \in \mathbb{Q}[x]$, we define $\gcd(f(x), g(x)) = d(x)$ to satisfy:

- $d(x) \mid f(x)$ and $d(x) \mid g(x)$
- For all $h(x) \in \mathbb{Q}[x]$ s.t. $h(x) \mid f(x)$ and $h(x) \mid g(x)$, $\|h(x)\| \leq \|d(x)\|$

Theorem 4 (Existence and uniqueness of gcd)**.** *Let $f(x), g(x) \in \mathbb{Q}[x]$, not both zero. Then there exists a unique monic polynomial $d(x) \in \mathbb{Q}[x]$ such that*

$$d(x) = \gcd(f(x), g(x)).$$

Proof. Let

$$S = \{\, a(x)f(x) + b(x)g(x) : a(x), b(x) \in \mathbb{Q}[x],\ a(x)f(x) + b(x)g(x) \neq 0 \,\}.$$

Since $f(x)$ and $g(x)$ are not both zero, the set S is nonempty. Also, every element of S has positive norm. So the set

$$N = \{\, \|s(x)\| : s(x) \in S \,\}$$

is a nonempty subset of $\mathbb{Z}^+$. By the well-ordering principle, N has a least element. Choose $d(x) \in S$ with smallest norm. Then for some $a(x), b(x) \in \mathbb{Q}[x]$,

$$d(x) = a(x)f(x) + b(x)g(x).$$

We first show that $d(x) \mid f(x)$. By the division algorithm, there exist $q(x), r(x) \in \mathbb{Q}[x]$ such that

$$f(x) = q(x)d(x) + r(x),$$

where either $r(x) = 0$ or

$$\|r(x)\| < \|d(x)\|.$$

Rearranging gives

$$r(x) = f(x) - q(x)d(x).$$

Since $d(x) = a(x)f(x) + b(x)g(x)$, we get

$$r(x) = f(x) - q(x)(a(x)f(x) + b(x)g(x)).$$

Thus

$$r(x) = (1 - q(x)a(x))f(x) + (-q(x)b(x))g(x).$$

So if $r(x) \neq 0$, then $r(x) \in S$, but it has smaller norm than $d(x)$, which contradicts how $d(x)$ was chosen. Therefore $r(x) = 0$, and hence

$$d(x) \mid f(x).$$

The same argument with $g(x)$ in place of $f(x)$ shows that

$$d(x) \mid g(x).$$

Now let $h(x) \in \mathbb{Q}[x]$ be any common divisor of $f(x)$ and $g(x)$. Then

$$f(x) = h(x)u(x) \qquad \text{and} \qquad g(x) = h(x)v(x)$$

for some $u(x), v(x) \in \mathbb{Q}[x]$. Hence

$$d(x) = a(x)f(x) + b(x)g(x) = a(x)h(x)u(x) + b(x)h(x)v(x).$$

Factoring gives

$$d(x) = h(x)(a(x)u(x) + b(x)v(x)).$$

Thus

$$h(x) \mid d(x).$$

So $d(x)$ satisfies the gcd property.

Finally, make $d(x)$ monic. Let $c \in \mathbb{Q}$ be the leading coefficient of $d(x)$. Since $d(x) \neq 0$, we have $c \neq 0$. Define

$$d_0(x) = c^{-1}d(x).$$

Then $d_0(x)$ is monic. Since c is a nonzero rational number, multiplying by c or c^{-1} does not change divisibility in $\mathbb{Q}[x]$. Therefore $d_0(x)$ is also a greatest common divisor of $f(x)$ and $g(x)$.

It remains to prove uniqueness. Suppose $d_1(x)$ and $d_2(x)$ are both monic greatest common divisors of $f(x)$ and $g(x)$. Since $d_1(x)$ is a gcd and $d_2(x)$ is a common divisor, we have

$$d_2(x) \mid d_1(x).$$

Similarly,

$$d_1(x) \mid d_2(x).$$

So there exist $u(x), v(x) \in \mathbb{Q}[x]$ such that

$$d_1(x) = d_2(x)u(x) \qquad \text{and} \qquad d_2(x) = d_1(x)v(x).$$

Taking degrees gives

$$\deg(d_1) = \deg(d_2) + \deg(u)$$

and

$$\deg(d_2) = \deg(d_1) + \deg(v).$$

Thus $\deg(u) = \deg(v) = 0$, so $u(x)$ and $v(x)$ are nonzero constant polynomials. Since $d_1(x)$ and $d_2(x)$ are both monic, the constant $u(x)$ must be 1. Therefore

$$d_1(x) = d_2(x).$$

Hence the monic greatest common divisor exists and is unique. $\square$

5. The Division Algorithm in $\mathbb{Q}[x]$

We need a definition of division in $\mathbb{Q}$[x] that holds similar properties to the integers. For $\mathbb{Q}[x]$, the norm plays the role that size plays in integer division. Since norms of nonzero polynomials are positive integers, we can use the well-ordering principle to pick a remainder of smallest norm.

Theorem 5 (Division algorithm in $\mathbb{Q}[x]$)**.** *Let $f(x), g(x) \in \mathbb{Q}[x]$ with $g(x) \neq 0$. Then there exist unique polynomials $q(x), r(x) \in \mathbb{Q}[x]$ such that $f(x) = q(x)g(x) + r(x)$, where either $r(x) = 0$ or $\|r(x)\| < \|g(x)\|$.*

Proof. **Existence.** Look at all possible remainders after subtracting multiples of g from f: $S = \{ f(x) - q(x)g(x) : q(x) \in \mathbb{Q}[x] \}$. If $0 \in S$, then $f = qg$ for some q, so $f = qg + 0$, and the result holds with $r = 0$.

Now suppose $0 \notin S$. Then every element of S is nonzero, so each element has a norm in $\mathbb{Z}^+$. Define $D = \{ \|s(x)\| : s(x) \in S \} \subseteq \mathbb{Z}^+$. The set D is nonempty because $f - 0g = f$ lies in S. By the well-ordering principle, D has a smallest element. Choose $r \in S$ with minimum norm. Since $r \in S$, there exists $q \in \mathbb{Q}[x]$ such that $r = f - qg$. Equivalently, $f = qg + r$.

It remains to show that $\|r(x)\| < \|g(x)\|$. Suppose instead that $\|r(x)\| \geq \|g(x)\|$. Write $r(x) = a_m x^m + \cdots$ and $g(x) = b_n x^n + \cdots$, where $a_m, b_n \neq 0$, $m = \deg(r)$, and $n = \deg(g)$. Since $\|r(x)\| \geq \|g(x)\|$ means $2^m \geq 2^n$, we get $m \geq n$. Since $b_n \neq 0$ and $b_n \in \mathbb{Q}$, the rational number a_m/b_n exists. Define $r^*(x) = r(x) - \frac{a_m}{b_n} x^{m-n} g(x)$.

The second term has leading term $a_m x^m$, so the leading term of r cancels. Thus either $r^* = 0$ or $\|r^*(x)\| < \|r(x)\|$. Also,

$$\begin{aligned} r^* &= f - qg - \frac{a_m}{b_n} x^{m-n} g \\ &= f - \left(q + \frac{a_m}{b_n} x^{m-n}\right) g. \end{aligned}$$

Therefore $r^* \in S$. If $r^* = 0$, then $0 \in S$, which is not the case we are in. If $r^* \neq 0$, then r^* is in S and has smaller norm than r, contradicting how we chose r. Therefore our assumption was false, and $\|r(x)\| < \|g(x)\|$.

Uniqueness. Suppose $f = q_1 g + r_1 = q_2 g + r_2$, where r_1 and r_2 are both either zero or have norm less than $\|g(x)\|$. Subtracting gives $g(q_1 - q_2) = r_2 - r_1$. Assume $q_1 \neq q_2$. Then $q_1 - q_2 \neq 0$, and since $\mathbb{Q}[x]$ has no zero divisors, $g(q_1 - q_2) \neq 0$. Hence

$$\begin{aligned} \|g(x)(q_1(x) - q_2(x))\| &= 2^{\deg(g(q_1 - q_2))} \\ &= 2^{\deg(g) + \deg(q_1 - q_2)} \\ &= \|g(x)\| \cdot \|q_1(x) - q_2(x)\| \\ &\geq \|g(x)\|. \end{aligned}$$

But $r_2 - r_1$ is either zero or has norm strictly less than $\|g(x)\|$, since each of r_1 and r_2 is either zero or has norm less than $\|g(x)\|$, so their degrees are both less than $\deg(g)$ when they are nonzero. This is impossible. Therefore $q_1 = q_2$. Substituting back gives $r_1 = r_2$. □

This result is extremely important as we now know we can "split" bigger polynomials into smaller ones. For example, take

$$3x^2 + 5x + 1 = (\frac{3}{2}x + 2)(2x + \frac{2}{3}) - \frac{2}{3}$$

Here, $3x^2 + 5x + 1$ is split into products of polynomials of degree 1, which is vital in future steps.

6. Euclidean Algorithm

The greatest common divisor of two polynomials is the same as the greatest common divisor of when you subtract multiples of one from another. This is the foundation of Euclid's Algorithm for rational polynomials.

Definition 13. Define **Euclidean Algorithm** on $f(x), g(x) \in \mathbb{Q}[\mathrm{x}]$ to produce $r, q \in \mathbb{Q}[\mathrm{x}]$ such that

$$f = q_1 g + r_1 \qquad g = q_2 r_1 + r_2 \qquad r_{n-2} = q_n r_{n-1} + r_n$$

Lemma 6.1. *For any $f(x), g(x) \in \mathbb{Q}[\mathrm{x}]$, the Euclidean Algorithm generates an $n \in \mathbb{Z}^+$ such that $r_n = 0$.*

Proof. Let $S = \{f(x) \in \mathbb{Q}[\mathrm{x}] \mid \exists\, g(x) \text{ s.t. } r_i \neq 0\ \forall i \in \mathbb{Z}\}$. For the sake of contradiction, S is not empty. By the well-ordering principle, there exists a least element l. Performing Euclid's Algorithm on (l, g), we have:

$$g = q_1 l + r_1$$
$$l = q_2 r_1 + r_2$$

If (g, l) does not reach 0, (l, r_1) will also not reach 0, as they generate the same values. Thus, $r_1 \in S$, but $r_l < l$ by the division algorithm, which is a contradiction. Thus, S must be empty. $\square$

Lemma 6.2. *For any $f(x), g(x) \in \mathbb{Q}[\mathrm{x}]$, if the Euclidean Algorithm generates $r_n = 0$, $r_{n-1} = \gcd(f, g)$.*

Proof. Let $S = \{f(x) \in \mathbb{Q}[\mathrm{x}] \mid \exists\, g(x) \text{ s.t. } r_{n-1} \neq \gcd(f, g)\}$. For the sake of contradiction, S is not empty. By the well-ordering principle, there exists a least element l. Performing Euclid's Algorithm on (l, g), we have:

$$g = q_1 l + r_1$$

If $\gcd(g, l) = d$, $d \mid g$ and $d \mid l$. Since $b > l$, $\gcd(l, r_1)$ remains d. However, the Euclidean Algorithm on (r_1, l) generates the same values as (b, l). Thus, $r_1 \in S$, but $r_l < l$ by the division algorithm, which is a contradiction. Thus, S must be empty. $\square$

Theorem 6. *Each remainder in Euclid's algorithm on $f(x), g(x) \in \mathbb{Q}[\mathrm{x}]$ can be expressed as a linear combination of $f(x)$ and $g(x)$, or*

$$a(x), b(x) \in \mathbb{Q}[\mathrm{x}] \qquad f(x)a_n(x) + g(x)b_n(x) = r_n(x)$$

Proof. Since $f = q_1 g + r_1$, $r_1 = f - q_1 g$. In addition, since $r_2 = g - q_2 r_1 = g - q_2 g + q_1 q_2 f$, r_2 can also be expressed as a linear combination of $f(x)$ and $g(x)$. Let

$$S := \left\{k \in \mathbb{Z}^+ \mid r_k \neq f(x)a(x) + g(x)b(x) \forall a(x), b(x) \in \mathbb{Q}[\mathrm{x}]\right\}$$

For the sake of contradiction, S is not empty. Thus, by the well-ordering principle, there exists a least element $l \in S$. However, $1, 2 \notin S$, so $l > 2$ and $l - 2, l - 1 \in \mathbb{Z}^+$. Since l is the least element of S, $l - 2, l - 1 \notin S$.

$$\exists a_1(x), b_1(x) \in \mathbb{Q}[\mathrm{x}] \qquad a_1 f + b_1 g = r_{l-1}$$
$$\exists a_2(x), b_2(x) \in \mathbb{Q}[\mathrm{x}] \qquad a_2 f + b_2 g = r_{l-2}$$

Thus, by the Euclidean Algorithm, we know that

$$\begin{aligned} r_{l-2} &= q_l r_{l-1} + r_l \\ a_2 f + b_2 g &= q_l(a_1 f + b_1 g) + r_l \\ r_l &= (a_2 - q_1 a_1) f + (b_2 - b_1) g \end{aligned}$$

As $(a_2 - q_1 a_1), (b_2 - b_1) \in \mathbb{Q}[\mathrm{x}]$, r_l can thus be written as a linear combination of $f(x)$ and $g(x)$. However, since $l \in S$, this is a contradiction and S must be empty.

$\square$

Corollary 7. *(Bezout's) The greatest common divisor of $f(x), g(x) \in \mathbb{Q}[\mathrm{x}]$ can be expressed as a linear combination of $f(x)$ and $g(x)$, or*

$$k_1(x), k_2(x) \in \mathbb{Q}[\mathrm{x}] \qquad f(x)k_1(x) + g(x)k_2(x) = \gcd(f(x), g(x))$$

Proof. With $r_n = 0$, the Euclidean algorithm always produces $r_{n-1} = \gcd(f(x), g(x))$. $\square$

7. Primes

7.1. Definition

Define $p(x) \in \mathbb{Q}[x]$ to be prime if:

- For all $a(x), b(x) \in \mathbb{Q}[x]$ where $a(x)b(x) = p(x)$, either $a(x)$ or $b(x)$ is a unit
- $p(x)$ is not a unit
- where $p(x) = \sum_0^n a_i x^i$, $a_n = 1$ (p(x) is monic)

Note that all monic linears $l(x) \in \mathbb{Q}[x]$ (polynomials of degree 1) are prime in $\mathbb{Q}[x]$ since they have a norm of 2. for any $a(x), b(x) \in \mathbb{Q}[x]$ where $a(x)b(x) = l(x)$, $\|a(x)\|\|b(x)\| = 2$ (norm preserves multiplication), and $\|a(x)\|, \|b(x)\| \in \mathbb{Z}^+$ (since they cannot be 0). Then, because 2 is prime in $\mathbb{Z}^+$, $\|a(x)\| = 1 \vee \|b(x)\| = 1$. This means that either a or b is a unit, and it follows then that $l(x)$ is a prime.

7.2. Prime Factors

Theorem 8 (Existence of prime factors)**.** *For all $a(x) \in \mathbb{Q}[x]$ where $a(x)$ isn't a unit, there exists a $p(x)$ which is prime where $p(x)|a(x)$.*

When $a(x) = 0$, it is divisible by all primes. For some polynomial $a(x)$ which isn't 0 or a unit, let $S = \{n(x) \in \mathbb{Q}[x] \mid n(x) \neq 0 \wedge \|n(x)\| \neq 1 \wedge n(x) \mid a(x)\}$. We know S is non-empty since $a(x) \in S$.

Then consider $\|S\| = \{\|n(x)\| \in S\}$. Since for all $n(x) \in \mathbb{Q}[x]$ where $n \neq 0$, $\|n(x)\| \in \mathbb{Z}^+$, $\|S\| \in \mathbb{Z}^+$. Then, since it is non-empty, there must exist an element $\|m(x)\| \in \|S\|$ which corresponds to $m(x) \in S$ where $\|m(x)\|$ is minimal in $\|S\|$.

Suppose there exists a $k(x) \mid m(x)$, where $\|k(x)\| \neq \|m(x)\|$ and isn't a unit. Since $\|k(x)\| \mid \|m(x)\|$ and $\|k(x)\|, \|m(x)\| \in \mathbb{Z}^+$, $\|k(x)\| \leq \|m(x)\|$, but $\|k(x)\| \neq \|m(x)\|$, so $\|k(x)\| < \|m(x)\|$. Since $k(x) \mid m(x)$, then because $m(x) \mid a(x)$, $k(x) \mid a(x)$, so $k(x) \in |S$. Then, we have $\|k(x)\| \in \|S\|$ too, but $\|k(x)\| < \|m(x)\|$, so $\|m(x)\|$ is not minimal in $S \rightarrow\leftarrow$.

Therefore, if there exists a $k(x) \mid m(x)$, $\|k(x)\| = \|m(x)\|$ or it is a unit. For any $a(x)b(x) = m(x)$, either $a(x)$ is a unit or $\|a(x)\| = \|m(x)\|$. In the latter case, this implies that $b(x)$ is a unit, since preservation of norm after multiplication is unique to units. Consequently, for all $a(x), b(x) \in \mathbb{Q}[x]$ where $a(x)b(x) = m(x)$, either $a(x)$ or $b(x)$ is a unit, and $m(x)$ isn't a unit, meaning that $m(x)$ is prime.

7.3. Prime Factorisation

Theorem 9 (Existence of a Prime Factorisation)**.** *For all $n(x) \in \mathbb{Q}[x]$, $n(x)$ can be expressed as $u(x)\prod_{i=1}^{n} p_i(x)^{\alpha_i}$ where $p_i(x) \in \mathbb{Q}[x]$ is prime, $a_i \in \mathbb{Z}^+$, and $u(x) \in \mathbb{Q}[x]^\times$.*

Let $S = \{(p_i, \alpha_i)\}$. For all $n(x) \in \mathbb{Z}^+, \exists m_1(x) \mid n(x)$ and $m_1(x)$ is prime. If $m_1(x) = n(x)$, then such S exists where $S = \{m_1\}$. Otherwise, there exists a $n_1(x)$ where $m_1(x)n_1(x) = n(x)$. It follows then that $\|m_1(x)\|\|n_1(x)\| = \|n(x)\|$, so $\|n_1(x)\| \mid \|n(x)\|$. Since they are both positive integers, this means that $\|n_1(x)\| \leq \|n(x)\|$. However, if $\|n_1(x)\| = \|n(x)\|$, then $\|m_1(x)\| = 1$, meaning it is a unit and therefore can't be prime. Therefore, $\|n_1(x)\| < \|n(x)\|$.

Similarly, either $n_1(x)$ is prime or $m_2(x)n_2(x) = n_1(x)$, where $m_2(x)$ is prime. If this terminates at $n_k(x)$ which is a unit, we have $n(x) = n_k(x)\prod_{i=1}^{k} m_i(x)$. Then, we can have $S' = \{\{m_i(x) \mid i \in \mathbb{Z} \wedge 0 < i \leq k\}\}$. Where $c(m_i(x))$ returns its multiplicity within S', we can have

$S = \{(m_i(x), c(m_i(x)) \mid m_i(x) \in S'\}$, which matches the properties required in our proposition.

Suppose this doesn't terminate. Then, we have set $\|N\| = \{\|n_i(x)\| | i \in \mathbb{Z}^+\}$ which is infinite, and for all $k \in \mathbb{Z}^+, \|n_k(x)\| > \|n_{k+1}(x)\|$. Also, $\|N\| \subset \mathbb{Z}^+$ since the norm of all non-zero polynomials is a positive integer. Since it is non-empty, there must be a minimal element $\|n_m(x)\|$ in it. However, $\|n_m(x)\| > \|n_{m+1}(x)\|$, so it isn't minimal $\rightarrow\leftarrow$.

Therefore, it must terminate, in which case there exists such set S.

8. Fundamental Lemma

The irreducibility of primes can be used by the following lemma.

Lemma 8.1 (Euclid's Lemma in $\mathbb{Q}[x]$)**.** *Let $p(x), f(x), g(x) \in \mathbb{Q}[x]$, and suppose $p(x)$ is prime. If $p(x) \mid f(x)g(x)$, then $p(x) \mid f(x)$ or $p(x) \mid g(x)$.*

Proof. Suppose $p(x) \mid f(x)g(x)$. If $p(x) \mid f(x)$, then we are done. So assume $p(x) \nmid f(x)$.

Let $d(x) = \gcd(p(x), f(x))$. Since $d(x) \mid p(x)$ and $p(x)$ is prime, $d(x)$ is either a unit or an associate of $p(x)$.

If $d(x)$ were an associate of $p(x)$, then $p(x) \mid d(x)$. Since $d(x) \mid f(x)$, this would imply $p(x) \mid f(x)$, contradicting our assumption. Therefore $d(x)$ is a unit.

Since gcds are chosen to be monic, the only monic unit is 1. Hence $d(x) = 1$. By Bezout's Identity, there exist $a(x), b(x) \in \mathbb{Q}[x]$ such that

$$a(x)p(x) + b(x)f(x) = 1.$$

Multiplying both sides by $g(x)$ gives

$$a(x)p(x)g(x) + b(x)f(x)g(x) = g(x).$$

Now $p(x)$ divides $a(x)p(x)g(x)$ because that term has a factor of $p(x)$. Also, $p(x)$ divides $b(x)f(x)g(x)$ because $p(x) \mid f(x)g(x)$. Therefore $p(x)$ divides their sum, so $p(x) \mid g(x)$.

Thus $p(x) \mid f(x)$ or $p(x) \mid g(x)$. $\square$

This helpful corollary extends this to the greatest common divisor and allows us to multiply them. If a number is relatively prime to two numbers, it must be relatively prime to their product.

Corollary 10. *If $f(x), g(x), h(x) \in \mathbb{Q}[\mathrm{x}]$, and $\gcd(f, g) = 1$ as well as $\gcd(f, h) = 1$ then $\gcd(f, gh) = 1$.*

Proof. Assume for the sake of contradiction that there exists $a(x) \in \mathbb{Q}[\mathrm{x}]$, $\|a(x)\| > 1$ such that $\gcd(f, gh) = a$. Thus, $a \mid f$ and $a \mid gh$. We know that $a(x)$ must have a prime factor. Define it as $p(x)$. Now, we know that

$$p(x) \mid f(x) \qquad p(x) \mid g(x)h(x)$$

By the Fundamental Lemma, $p(x) \mid g(x)$ or $p(x) \mid h(x)$. Without loss of generality, let $p(x) \mid g(x)$. Thus, since $p(x) \mid f(x)$, $\gcd(f(x), g(x)) \geq \|p(x)\| > 1$. However, this is a contradiction. Thus, $\gcd(f, gh) = 1$ $\square$

Now, we can extend these lemmas to a more general case that proves if a number divides a product, not all numbers in the product can be relatively prime to that number.

Theorem 11. *Let $f(x), g_i(x) \in \mathbb{Q}[x]$, and let $f(x) \mid \prod_{i=1}^{n} g_i(x)$. Then, there exists $k \leq n$ such that $\gcd(f(x), g_k(x)) \neq 1$.*

Proof. Prove that if $\gcd(f(x), g_i(x)) = 1$ for all $i \leq n$, then $\gcd(f(x), \prod_{i=1}^{n} g_i(x)) = 1$. Let $S = j \in \mathbb{Z}^+ | \gcd(f(x), \prod_{i=1}^{n} g_i(x)) \neq 1.$. For the sake of contradiction, S is not empty. Then, since $S \subseteq \mathbb{Z}^+$, by the well-ordering principle, there exists a least element $l \in S$. Note that if $j = 1$, then

$$\gcd(f(x), \prod_{i=1}^{1} g_i(x)) = \gcd(f(x), g_1(x)) = 1$$

which is a contradiction. Thus, since $1 \notin S$, $l > 1$ and $l - 1 \notin S$. So, we know that

$$\gcd(f(x), \prod_{i=1}^{l-1} g_i(x)) = 1$$

Since $\gcd(f(x), \prod_{i=1}^{l-1} g_i(x)) = 1$ and $\gcd(f(x), g_l(x)) = 1$, $\gcd(f(x), g_l(x) \prod_{i=1}^{l-1} g_i(x)) = \gcd(f(x), \prod_{i=1}^{l} g_i(x)) = 1$. However, since $l \in S$, this is a contradiction. Thus, S must be empty. As $\gcd(f(x), \prod_{i=1}^{n} g_i(x)) = 1$ is a contradiction to the fact that $f(x) \mid \prod_{i=1}^{n} g_i(x)$, $\gcd(f(x), g_i(x))$ cannot equal 1 for all $i \leq n$. $\square$

To deal with powers of primes in prime facorizations, we need one more lemma that proves powers of primes do not divide each other.

Lemma 8.2. $\gcd(p^k, q^j) = 1$ *for all* $p, q \in \mathbb{Q}[\mathrm{x}]$ *such that* $\gcd(p, q) = 1$.

Proof. First, note that if $k = 0$ or $j = 0$, $\gcd(f(x), 1) = 1 \forall f(x) \in \mathbb{Q}[\mathrm{x}]$. Next, prove that $\gcd(p, q^j) = 1$. Let $S = \left\{i \in \mathbb{Z}^+ \mid \gcd(p, q^i) \neq 1\right\}$ For the sake of contradiction, let S be nonempty. By the well-ordering principle, there exists a least element $l \in S$. Note that if $i = 1$, $\gcd(p, q^1) =$

$\gcd(p, q) = 1$. Thus, $1 \notin S$ and $l > 1$, and $l - 1 \notin S$.

$$\begin{aligned} \gcd(p, q^{l-1}) &= 1 \\ \gcd(p, q) = 1 \gcd(p, q^l) &= 1 \end{aligned}$$

Thus, this is a contradiction and S is empty. An analogous argument can be made since $\gcd(p, q^j) = 1$ proving that $\gcd(p^k, q^j) = 1$. □

Finally, we can prove the uniqueness of prime factorizations. For a prime and its power to not exist in another facorization, either that facoritzation doesnt include that prime or it includes it but with a different power. Here, we prove (using the Fundamental Lemma) that both cases are impossible.

Lemma 8.3. *If $f(x) \in \mathbb{Q}[\mathrm{x}]$ can be expressed as $u(x) \prod_{i=1}^{n} p_i(x)^{\alpha_i}$ where $p_i(x) \in \mathbb{Q}[\mathrm{x}]$ is prime and unique, $u(x) \in \mathbb{Q}[\mathrm{x}]^\times$, $\alpha_i \in \mathbb{Z}^+$, then only the set $S = \{(p_i, \alpha_i)\}$ has this property.*

Proof. For the sake of contradiction, there exists $A = \{(q_i, \beta_i)\}$ of size m, s.t.

$$f(x) = u(x) \prod_{i=1}^{m} q_i(x)^{\beta_i}$$

and there exists $(q_k, \beta_k) \in A$ s.t. $(q_k, \beta_k) \notin S$. This can be true in two ways.
Case 1: For all $\phi \in \mathbb{Z}^+$, $(q_k(x), \phi) \notin S$.

$$\begin{aligned} u(x) \prod_{i=1}^{m} q_i(x)^{\beta_i} &= u(x) \prod_{i=1}^{n} p_i(x)^{\alpha_i} \\ \prod_{i=1}^{m} q_i(x)^{\beta_i} &= \prod_{i=1}^{n} p_i(x)^{\alpha_i} \\ \prod_{i=1}^{k-1} q_i(x)^{\beta_i} \cdot q_k^{\beta_k} \cdot \prod_{i=k+1}^{m} q_i(x)^{\beta_i} &= \prod_{i=1}^{n} p_i(x)^{\alpha_i} \end{aligned}$$

Thus, we know that $q_k(x)^{\beta_k} \mid \prod_{i=1}^{n} p_i(x)^{\alpha_i}$. By Lemma ???? we know that $q_k^{\beta_k} \mid p_i(x)^{\alpha_i}$ for some i, yet since $p_i \neq q_k$, we know that

$$\forall p_i \in S \quad \gcd(p_i^{\alpha_i}, q_k^{\beta_k}) = 1 \qquad \text{Lemma ??????}$$

Thus, this is a contradiction.
Case 2: For all $a(x) \in \mathbb{Q}[\mathrm{x}]$, $(a(x), \beta_k) \notin S$. From the above, we know that there exists

$(p_j(x), \alpha_j) \in S$ such that $p_j(x) = q_k(x)$. Again,

$$\begin{aligned}
u(x)\prod_{i=1}^{m} q_i(x)^{\beta_i} &= u(x)\prod_{i=1}^{n} p_i(x)^{\alpha_i}\\
\prod_{i=1}^{m} q_i(x)^{\beta_i} &= \prod_{i=1}^{n} p_i(x)^{\alpha_i}\\
\prod_{i=1}^{k-1} q_i(x)^{\beta_i}\cdot q_k^{\beta_k}\cdot \prod_{i=k+1}^{m} q_i(x)^{\beta_i} &= \prod_{i=1}^{j-1} p_i(x)^{\alpha_i}\cdot q_j^{\alpha_j}\cdot \prod_{i=j+1}^{n} p_i(x)^{\alpha_i}
\end{aligned}$$

Now, we have 2 cases, as $\beta_k \neq \alpha_j$.
Case 2.1: $\beta_k > \alpha_j$

$$\begin{aligned}
\prod_{i=1}^{k-1} q_i(x)^{\beta_i}\cdot q_k^{\beta_k}\cdot \prod_{i=k+1}^{m} q_i(x)^{\beta_i} &= \prod_{i=1}^{j-1} p_i(x)^{\alpha_i}\cdot q_j^{\alpha_j}\cdot \prod_{i=j+1}^{n} p_i(x)^{\alpha_i}\\
q_k^{\beta_k-\alpha_j}\cdot q_k^{\beta_k}\prod_{i=1}^{k-1} q_i(x)^{\beta_i}\cdot \prod_{i=k+1}^{m} q_i(x)^{\beta_i} &= q_k^{\beta_k-\alpha_j}\cdot q_j^{\alpha_j}\cdot \prod_{i=1}^{j-1} p_i(x)^{\alpha_i}\cdot \prod_{i=j+1}^{n} p_i(x)^{\alpha_i}\\
q_k^{\beta_k}\cdot q_k^{\beta_k-\alpha_j}\cdot \prod_{i=1}^{k-1} q_i(x)^{\beta_i}\cdot \prod_{i=k+1}^{m} q_i(x)^{\beta_i} &= q_k^{\beta_k}\cdot \prod_{i=1}^{j-1} p_i(x)^{\alpha_i}\cdot \prod_{i=j+1}^{n} p_i(x)^{\alpha_i}\\
q_k^{\beta_k-\alpha_j}\cdot \prod_{i=1}^{k-1} q_i(x)^{\beta_i}\cdot \prod_{i=k+1}^{m} q_i(x)^{\beta_i} &= \prod_{i=1}^{j-1} p_i(x)^{\alpha_i}\cdot \prod_{i=j+1}^{n} p_i(x)^{\alpha_i}
\end{aligned}$$

Thus, $q_k^{\beta_k-\alpha_j} \mid \prod_{i=1}^{j-1} p_i(x)^{\alpha_i}$ or $q_k^{\beta_k-\alpha_j} \mid \prod_{i=j+1}^{n} p_i(x)^{\alpha_i}$ by the Fundamental Lemma. However, neither can be true as for all $p_i \in S \quad \gcd(p_i^{\alpha_i}, q_k^{\beta_k}) = 1$. Thus, this is a contradiction.
Case 2.2: $\beta_k < \alpha_j$ can be proven using an analogous argument. □

9. Conclusion

"Mathematics consists in proving the most obvious thing in the least obvious way."

– George Pólya

Finally, we have proved that every number has a unique factorization. This can be formalized in our final theorem.

Theorem 12. *(Unique Factorization) Every rational polynomial* $f(x) \in \mathbb{Q}[\text{x}]$ *can be expressed as* $u(x)\prod_{i=1}^{n} p_i(x)^{\alpha_i}$ *where the set* $S = \{(p_i, \alpha_i)\}$ *is unique.*

This fundamental result is not possible for the simple integer polynomial ring, $\mathbb{Z}[x]$. It is true for $\mathbb{Q}[x]$ since $\mathbb{Q}$ has the property that every number has an inverse. However, the notion of "primes" still remains as some polynomials, no matter what degree, cannot be factored.

Appendix A: Summation and Product Lemmas

Lemma 9.1. $\sum\limits_{i=i_0}^{n} a_i + \sum\limits_{i=i_0}^{n} b_i = \sum\limits_{i=i_0}^{n} (a_i + b_i)$

Proof. Note that

$$\sum_{i=i_0}^{i_0} a_i + \sum_{i=i_0}^{i_0} b_i = a_{i_0} + b_{i_0} = \sum_{i=i_0}^{i_0} (a_i + b_i)$$

Let $S = \left\{ k \in \mathbb{Z}^+ \;\middle|\; \sum\limits_{i=i_0}^{i_0+k} a_i + \sum\limits_{i=i_0}^{i_0+k} b_i \neq \sum\limits_{i=i_0}^{i_0+k} (a_i + b_i) \right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.
If $k = 1$:

$$\sum_{i=i_0}^{i_0+1} a_i + \sum_{i=i_0}^{i_0+1} b_i = a_{i_0+1} + a_{i_0} + b_{i_0+1} + b_{i_0} = (a_{i_0+1} + b_{i_0+1}) + (a_{i_0} + b_{i_0}) = \sum_{i=i_0}^{i_0+1} (a_i + b_i)$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i = \sum_{i=i_0}^{i_0+l-1} (a_i + b_i)$$

$$a_l + b_l + \sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i = a_l + b_l + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i)$$

$$a_l + \sum_{i=i_0}^{i_0+l-1} a_i + b_l + \sum_{i=i_0}^{i_0+l-1} b_i = (a_l + b_l) + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i)$$

$$\sum_{i=i_0}^{i_0+l} a_i + \sum_{i=i_0}^{i_0+l} b_i = \sum_{i=i_0}^{i_0+l} (a_i + b_i)$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. ∎

Lemma 9.2. *Sum Constants:* $\sum\limits_{i=i_0}^{n} ca_i = c \sum\limits_{i=i_0}^{n} a_i$

Proof. Note that

$$\sum_{i=i_0}^{i_0} ca_i = ca_{i_0} = c\sum_{i=i_0}^{i_0} a_i$$

and that

$$\sum_{i=i_0}^{i_0+1} ca_i = ca_{i_0} + ca_{i_0+1} = c\sum_{i=i_0}^{i_0+1} a_i$$

Let $S = \left\{k \in \mathbb{Z}^+ \;\middle|\; \sum_{i=i_0}^{i_0+k} ca_i \neq c\sum_{i=i_0}^{i_0+k} a_i\right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the well-ordering principle, there exists a least element $l \in S$. Since $1 \notin S$, $l > 1$ by NIBZO. As $l-1 \in \mathbb{Z}$ yet $l-1 \notin S$,

$$\sum_{i=i_0}^{i_0+l-1} ca_i = c\sum_{i=i_0}^{i_0+l-1} a_i$$
$$ca_l + \sum_{i=i_0}^{i_0+l-1} ca_i = ca_l + c\sum_{i=i_0}^{i_0+l-1} a_i$$
$$\sum_{i=i_0}^{i_0+l} ca_i = c\sum_{i=i_0}^{i_0+l} a_i$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Lemma 9.3. *Multiplied Sums:* $\sum_{i=i_0}^{n} a_i \cdot \sum_{j=j_0}^{m} b_j = \sum_{i=i_0}^{n}\sum_{j=j_0}^{m} (a_i b_j)$

Proof. By Sum Constants, we know that

$$\sum_{i=i_0}^{n} a_i \cdot \sum_{j=j_0}^{m} b_j = \sum_{i=i_0}^{n}(a_i \cdot \sum_{j=j_0}^{m} b_j)$$
$$\sum_{i=i_0}^{n} a_i \cdot \sum_{j=j_0}^{m} b_j = \sum_{i=i_0}^{n}(\sum_{j=j_0}^{m} a_i b_j)$$

$\square$

Lemma 9.4. $\prod_{i=i_0}^{n} a_i \prod_{i=i_0}^{n} b_i = \prod_{i=i_0}^{n} a_i b_i$

Proof. Note that

$$\prod_{i=i_0}^{i_0} a_i \prod_{i=i_0}^{i_0} b_i = a_{i_0} b_{i_0} = \prod_{i=i_0}^{i_0} a_i b_i$$

Let $S = \left\{k \in \mathbb{Z}^+ \;\middle|\; \prod_{i=i_0}^{i_0+k} a_i \prod_{i=i_0}^{i_0+k} b_i \neq \prod_{i=i_0}^{i_0+k} a_i b_i\right\}$. For the sake of contradiction, S is not empty. As

$S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.
If $k = 1$:

$$\prod_{i=i_0}^{i_0+1} a_i \prod_{i=i_0}^{i_0+1} b_i = a_{i_0+1}a_{i_0} \cdot b_{i_0+1}b_{i_0} = (a_{i_0+1}b_{i_0+1})(a_{i_0}b_{i_0}) = \prod_{i=i_0}^{i_0+1} a_i b_i$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i = \prod_{i=i_0}^{i_0+l-1} a_i b_i$$
$$a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i = a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i b_i$$
$$(a_l \prod_{i=i_0}^{i_0+l-1} a_i) \cdot (b_l \prod_{i=i_0}^{i_0+l-1} b_i) = (a_l b_l) \prod_{i=i_0}^{i_0+l-1} a_i b_i$$
$$\prod_{i=i_0}^{i_0+l} a_i \prod_{i=i_0}^{i_0+l} b_i = \prod_{i=i_0}^{i_0+l} a_i b_i$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. □

Lemma 9.5. $\prod_{i=i_0}^{n} a_i = \prod_{i=i_0}^{k-1} a_i \cdot a_k \cdot \prod_{i=k+1}^{n} a_i$

Proof. Let $S = \left\{ t \in \mathbb{Z}^+ \;\middle|\; \prod_{i=i_0}^{k+t} a_i \neq \prod_{i=i_0}^{k-1} a_i \cdot a_k \cdot \prod_{i=k+1}^{k+t} a_i \right\}$. We show S is empty. First, $1 \notin S$, since by the recursive definition of product,

$$\prod_{i=i_0}^{k+1} a_i = a_{k+1} \cdot \prod_{i=i_0}^{k} a_i = a_{k+1} \cdot a_k \cdot \prod_{i=i_0}^{k-1} a_i = \prod_{i=i_0}^{k-1} a_i \cdot a_k \cdot \prod_{i=k+1}^{k+1} a_i.$$

For the sake of contradiction, suppose S is not empty. Since $S \subseteq \mathbb{Z}^+$, by WOP there is a least element $l \in S$. Since $1 \notin S$, $l > 1$, so $l - 1 \in \mathbb{Z}^+$. Since l is least in S, $l - 1 \notin S$. Thus

$\prod_{i=i_0}^{k+l-1} a_i = \prod_{i=i_0}^{k-1} a_i \cdot a_k \cdot \prod_{i=k+1}^{k+l-1} a_i$. Then, using the recursive definition again,

$$\begin{aligned}
\prod_{i=i_0}^{k+l} a_i &= a_{k+l} \cdot \prod_{i=i_0}^{k+l-1} a_i \\
&= a_{k+l} \cdot \left(\prod_{i=i_0}^{k-1} a_i \cdot a_k \cdot \prod_{i=k+1}^{k+l-1} a_i \right) \\
&= \prod_{i=i_0}^{k-1} a_i \cdot a_k \cdot \prod_{i=k+1}^{k+l} a_i.
\end{aligned}$$

So $l \notin S$, a contradiction. Therefore S is empty. Since $n > k$, write $n = k + t$ for some $t \in \mathbb{Z}^+$. Since S is empty, the equality holds for n. □

Appendix B: Exponent Lemmas

Lemma 9.6. $a^n a^m = a^{n+m} \quad \forall a \in R,\ n, m \in \mathbb{Z}^+ \cup \{0\}$

Proof. Let $S = \{n \in \mathbb{Z}^+ \mid \exists m \in \mathbb{Z}^+ \cup \{0\} \text{ s.t. } a^n a^m \neq a^{n+m}\}$. For the sake of contradiction, S is not empty. By the well-ordering principle, since $S \subset \mathbb{Z}^+ \cup \{0\}$, there must be a least element $l \in S$. First, for all $m \in \mathbb{Z}^+ \cup \{0\}$,

$$a^0 \cdot a^m = a^m$$

Thus, $0 \notin S$. Therefore, by NIBZO $l > 0$ and $l \in \mathbb{Z}^+$. Then, $l - 1 \in \mathbb{Z}^+ \cup \{0\}$ but $l - 1 \notin S$ since l is minimal.

$$\begin{aligned}
a^{l-1} a^m &= a^{l+m-1} \\
a \cdot a^{l-1} a^m &= a \cdot a^{l+m-1} \\
a^l a^m &= a^{l+m}
\end{aligned}$$

Thus, $l \notin S \rightarrow\leftarrow$. Then, S must be empty. □